

ADAM ZAATAR

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EDUCATION

Bowdoin College

B.A., Computer Science and Economics, Minor in German

Brunswick, ME

Expected May 2027 · GPA 3.53

Relevant coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Computer Systems, Econometrics, Artificial Intelligence

Learning Assistant, CSCI 1101 (Intro to CS), Bowdoin CS Department — Fall 2026

EXPERIENCE

Software Engineering Intern, ProgressSoft Corporation

Amman, Jordan · Jun–Aug 2026

- Built EventGuard at a payments software company: a seven-module Java 21 and Spring Boot service that ingests payment files, parses and validates records, detects duplicate payment IDs, and applies typed row-level rejection handling that separates parser errors from business-rule failures. github.com/adamzatar/EventGuard
- Structured the system as ports and adapters, keeping the core domain free of framework dependencies and assembling concrete implementations only at the CLI and Spring composition roots, so storage or transport can change without touching business logic.
- Implemented transactional persistence to PostgreSQL so a failed import cannot leave a partially written batch: explicit commit and rollback with suppressed-exception handling, try-with-resources on every connection, statement, and result set, and monetary values held as BigDecimal against a NUMERIC column to avoid floating-point drift.
- Kept a complete audit trail of every import: rejected rows are preserved with a typed reason rather than discarded, and the upload API returns structured errors across malformed requests, unsupported files, and persistence and I/O failures. Verified with 107 passing tests including a live database integration test, and changed the design safely through merge requests and twice-weekly mentor code review.

PROJECTS

Operating Systems (CSCI 3310)

C / C++ · team of two

- Built a user-level thread library: ucontext context switching, FIFO scheduling, locks and condition variables under Mesa semantics, and atomicity through interrupt masking, validated with a test suite written to expose buggy implementations.
- Implemented an external virtual-memory pager: demand paging, second-chance (clock) page replacement, software-tracked reference and dirty bits, and disk swap, with work deferral tuned to the cost gap between faults, zero-fills, and disk I/O.

ClickErase

Python, Gradio, MobileSAM, LaMa, Hugging Face Spaces

- Built and deployed a public image-editing app on my own: click an object, generate a segmentation mask with MobileSAM, and remove it with LaMa inpainting. Live demo and github.com/adamzatar/clickerase.

RESEARCH

- AI Layoffs and Investor Reactions (advised by Prof. Botsch): hand-coded events, cross-checked SEC filings, and estimated abnormal returns via regression. Gibbons Summer Research Fellowship (funded, advised by Prof. Erik Nelson): econometric modeling and experimental design.

SKILLS

Languages: Java, Python, JavaScript/TypeScript, SQL, C++, C

Backend & Web: Spring Boot, Spring MVC, REST API design, JDBC, transactional persistence, React, Next.js, Node.js

Testing & Practices: test-driven development (JUnit 5, Mockito), integration testing, code review, Git/GitLab, CI/CD (GitHub Actions)

AI-Assisted Development: scaffolding, unfamiliar APIs, and test data; generated code validated against tests written first and reviewed line by line before it ships

Systems & Fundamentals: data structures & algorithms, concurrency and synchronization, memory management, distributed systems, Linux/POSIX, Docker

Storage & Data: PostgreSQL, Redis, MySQL, SQLite, schema design, transactions

Spoken Languages: Arabic (native), English (fluent), German (intermediate)

LEADERSHIP

- Founder and President, Bowdoin Martial Arts Club (secured \$800 in funding, grew membership to over 30); Student Advisor, Bowdoin CS Department faculty hiring.